

1. main.c

```
#include "header.h"

#include <stdio.h>

#include <ctype.h>

int main(){

    int questionNumber;

    printf("Enter Question Number ( 1 - 15 ):");

    scanf("%d",&questionNumber);

    getchar();

    switch (questionNumber){

        case 1: {

            ThreeNumbersInLine(10);

            break;

        }

        case 2: {

            EvenNumbersCount(1 , 200);

            break;

        }

        case 3: {

            char str[1000];

            printf("Enter text: ");

            scanf("%[^\n]s",str);

            NumberOfVowels(str , 1000);

            break;

        }

        case 4:{

            float a , b;

            printf("Enter two floating point numbers: ");

            scanf("%f%f", &a, &b);

            int sum = AddTwoFloat(a, b);

            printf("Sum of %f and %f converted to int is %d", a, b, sum);
```

```

        break;
    }
case 5:{
    float a;

    printf("Enter floating point number: ");

    scanf("%f", &a);

    printf("Rightmost digit of integral part: %d", RightMost(a));

    break;
}
case 6:{
    float principal, rate, time;

    printf("Enter the principal amount: ");

    scanf("%f", &principal);

    printf("Enter the rate of interest (in percentage): ");

    scanf("%f", &rate);

    printf("Enter the time period (in years): ");

    scanf("%f", &time);

    printf("Simple Interest = %.2f\n", SimpleInterest(principal , rate , time));

    printf("Compound Interest = %.2f\n", CompoundInterest(principal , rate , time));

    break;
}
case 7:{
    const float HRA = 10.0 / 100;

    const float TA = 5.0 / 100;

    float base, salary;

    printf("Enter the basic salary : ");

    scanf("%f",&base);

    salary = base + (base * HRA) + (base * TA);

    printf("The salary of the employee is : %f",salary);

    break;
}

```

```

case 8:{
    bill();
    break;
}

case 9:{
    int number;
    printf("Enter an integer: ");
    scanf("%d", &number);
    printf("Decimal: %d\n", number);
    printf("Octal: %o\n", number);
    printf("Hexadecimal: %x\n", number);
    break;
}

case 10:{
    float value;
    printf("Enter a floating point value: ");
    scanf("%f", &value);
    printf("Value in exponent form (correct to two decimal places): %.2e\n", value);
    printf("Value in exponent form (correct to four decimal places): %.4e\n", value);
    printf("Value in exponent form (correct to eight decimal places): %.8e\n", value);
    break;
}

case 11:{
    char ch;
    printf("Enter character : ");
    scanf("%c", &ch);
    if(ch >= 'a' && ch <= 'z'){
        printf("Captial letter is: %c", ch - 32);
    }
    else if(ch >= 'A' && ch <= 'Z'){
        printf("Small letter is: %c", ch + 32);
    }
}

```

```

    }
    break;
}
case 12:{
    float a,b,c;
    float result;
    printf("Enter three floating point numbers: ");
    scanf("%f%f%f", &a, &b, &c);
    result = a + b + c;
    printf("Their sum is %.2f",result);
    break;
}
case 13:{
    char s;
    printf("Enter a character: ");
    scanf("%c",&s);
    if(s>='a' && s <='z'){
        printf("It is lower-case alphabet");
    }
    else if(s>='A' && s <='Z'){
        printf("It is upper-case alphabet");
    }
    else if(s>='0' && s<='9'){
        printf("It is Number");
    }
    else{
        printf("It is Special Character");
    }
    break;
}
case 14:{

```

```

int n;

printf("Enter the value of n: ");

scanf("%d", &n);

int sum = 0;

float avg;

for(int i = 1; i <= n; i++){

    sum += i;

}

avg = sum / n;

printf("The sum is: %d\n", sum);

printf("The average is: %f\n", avg);

break;

}

case 15:{

    int m, n, i;

    printf("Enter the values of m and n: ");

    scanf("%d%d", &m, &n);

    printf("Odd Numbers are: ");

    for (i = m; i < n; i++){

        if (i % 2 == 1)

            printf("%d ", i);

    }

    break;

}

case 16:{

    int m, n, i, j;

    printf("Enter the range m and n : ");

    scanf("%d %d", &m, &n);

    printf("Prime Numbers are: ");

    for(i = m; i <= n; i++){

        int flag = 0;

```

```

    for(j = 2; j <= i - 1; j++){
        if(i % j == 0){
            flag = 1;
            break;
        }
    }
    if(i == 1){
        continue;
    }
    else if(flag == 0){
        printf("%d ", i);
    }
    else{
        continue;
    }
}
break;
}
case 17:{
    int a;
    printf("Enter Numbers (enter -1 to stop):\n");
    while(1){
        scanf("%d", &a);
        if(a == -1){
            break;
        }
        Armstrong(a);
    }
    break;
}
case 18:{

```

```

int wind;

float temp;

float WCI;

printf("Enter the wind speed:");

scanf("%d",&wind);

getchar();

printf("Enter the temperature:");

scanf("%f", &temp);

if(wind >= 0 && wind <= 4){

    WCI = temp;

    printf("The value of WCI is: %f.\n", WCI);

}

else if(wind >= 45){

    WCI = 1.6 * temp - 55;

    printf("The value of WCI is: %f.\n",WCI);

}

else{

    WCI=91.4+(91.4-temp)*(0.0203*wind-0.304*wind*1/2-0.474);

    printf("The value of WCI is: %f.\n",WCI);

}

break;

}

case 19:{

    int n;

    printf("Enter an integer: ");

    scanf("%d", &n);

    printf("Is %d divisible by 5 and 6? ", n);

    if ((n % 5 == 0) && (n % 6 == 0)){

        printf("true\n");

    } else {

        printf("false\n");

    }

}

```

```
}

printf("Is %d divisible by 5 or 6? ", n);
if ((n % 5 == 0) || (n % 6 == 0)){
    printf("true\n");
} else {
    printf("false\n");
}

printf("Is %d divisible by 5 or 6, but not both? ", n);
if (((n % 5 == 0) || (n % 6 == 0)) && ((n % 5 != 0) && (n % 6 != 0))){
    printf("true\n");
} else {
    printf("false\n");
}

break;
}

case 20:{
    bill2();
    break;
}

default:
    break;
}

return 0;
}
```

2. header.h

```
#define MAX_ITEMS 100
```

```
typedef struct {  
    char name[50];  
    int quantity;  
    float pricePerUnit;  
} Item;
```

```
typedef struct{  
    char name[50];  
    float price;  
}Item1;
```

```
void ThreeNumbersInLine(int n);  
void EvenNumbersCount(int min , int max);  
void NumberOfVowels(char str[] , int size);  
int AddTwoFloat(float a , float b);  
int RightMost(float a);  
float SimpleInterest(float p , float r , float t);  
float CompoundInterest(float p , float r , float t);  
void bill();  
void Armstrong(int n);  
void bill2();
```

3. logic.c

```
#include "header.h"
```

```
#include <stdio.h>
```

```
#include <limits.h>
```

```
#include <stdlib.h>
```

```
#include <math.h>
```

```
void ThreeNumbersInLine(int n){
```

```
    int numbers[n];
```

```
    printf("Enter %d Numbers : \n", n);
```

```
    for(int i = 0 ; i < n ; i++){
```

```
        scanf("%d",&numbers[i]);
```

```
    }
```

```
    for(int i = 0 ; i < n ; i++){
```

```
        if(i % 3 == 2){
```

```
            printf("%d", numbers[i]);
```

```
            printf("\n");
```

```
        }
```

```
    else{
```

```
        printf("%d, ", numbers[i]);
```

```
    }
```

```
}
```

```
}
```

```
void EvenNumbersCount(int min , int max){
```

```
    int count = 0;
```

```
    int sum = 0;
```

```
    for(int i = min ; i <= max ; i++){
```

```
        if(i % 2 == 0){
```

```
            count++;
```

```
            sum += i;
```

```

    }
}

printf("Even Numbers between %d to %d are %d and their sum is : %d", min, max, count, sum);
}

```

```

void NumberOfVowels(char str[] , int size){

    int count = 0;

    for(int i = 0 ; str[i] != '\0' ; i++){

        if((str[i] == 'a') || (str[i] == 'e') || (str[i] == 'i') || (str[i] == 'o') || (str[i] == 'u') || (str[i] == 'A') ||
(str[i] == 'E') || (str[i] == 'I') || (str[i] == 'O') || (str[i] == 'U')){

            count++;

        }

    }

    printf("Number of Vowels are %d", count);

}

```

```

int AddTwoFloat(float a , float b){

    float sum = a + b;

    int s = (int)sum;

    return s;

}

```

```

int RightMost(float a){

    int s = (int)a;

    int r = s % 10;

    return r;

}

```

```

float SimpleInterest(float p , float r , float t){

    float si = (p * r * t)/100;

    return si;

}

```

```
}
```

```
float CompoundInterest(float p , float r , float t){
```

```
    float ci = p * pow((1 + r / 100) , t) - p;
```

```
    return ci;
```

```
}
```

```
void bill(){
```

```
    Item items[MAX_ITEMS];
```

```
    int n;
```

```
    float totalAmount = 0.0;
```

```
    printf("Enter the number of items: ");
```

```
    scanf("%d", &n);
```

```
    if (n <= 0 || n > MAX_ITEMS) {
```

```
        printf("Invalid number of items.\n");
```

```
        return;
```

```
    }
```

```
    for (int i = 0; i < n; i++) {
```

```
        printf("\nEnter details for item %d:\n", i + 1);
```

```
        printf("Item name: ");
```

```
        scanf(" %[^\\n]", items[i].name);
```

```
        printf("Quantity: ");
```

```
        scanf("%d", &items[i].quantity);
```

```
        printf("Price per unit: ");
```

```
        scanf("%f", &items[i].pricePerUnit);
```

```
    }
```

```
    printf("\n***** B I L L *****\n");
```

```
    printf("ITEM\t\tQUANTITY\t\tPRICE\t\tAMOUNT\n");
```

```
printf("_____\\n");
```

```
for (int i = 0; i < n; i++) {  
    float amount = items[i].quantity * items[i].pricePerUnit;  
    printf("%-15s\\t%d\\t\\t%.2f\\t\\t%.2f\\n", items[i].name, items[i].quantity, items[i].pricePerUnit,  
amount);  
    totalAmount += amount;  
}
```

```
printf("-----\\n");  
printf("Total Amount: %.2f\\n", totalAmount);  
printf("-----\\n");  
}
```

```
void Armstrong(int n){  
    int original, power, lastdigit, sum = 0, count = 0;  
    original = n;  
    while(n != 0){  
        n = n / 10;  
        count++;  
    }  
    n = original;
```

```
while(n != 0){  
    lastdigit = n % 10;  
    power = 1;  
    for(int i = 0 ; i < count ; i++){  
        power *= lastdigit;  
    }  
    sum = sum + power;  
    n = n / 10;
```

```

}

if(sum == original){

    printf("The Number %d is an Armstrong Number\n", original);

}

else{

    printf("The Number %d is not an Armstrong Number\n", original);

}

}

```

```

void bill2(){

    Item1 item[MAX_ITEMS];

    int n , shipping;

    float Amount , totalAmount , ShippingAmount;

    printf("Enter the number of items : ");

    scanf("%d",&n);

    for(int i=0;i<n;i++){

        printf("Enter the item : ");

        scanf("%s", item[i].name);

        printf("Enter the price : ");

        scanf("%f",&item[i].price);

        Amount += item[i].price;

    }

    printf("Overnight delivery (Enter 0 for no and 1 for yes): ");

    scanf("%d",&shipping);


    if(Amount > 100 && shipping == 1){

        ShippingAmount = 80;

        totalAmount = Amount + 80;

    }

    else if(Amount > 100 && shipping == 0){

        ShippingAmount = 30;

```

```
        totalAmount = Amount + 30;
    }
    else if(Amount < 100 && shipping == 1){
        ShippingAmount = 70;
        totalAmount = Amount + 70;
    }
    else{
        ShippingAmount = 20;
        totalAmount = Amount + 20;
    }
    printf(":Invoice\n");
    for(int i = 0 ; i < n ; i++){
        printf("%-15s ",item[i].name);
        printf("Rs. %f \n",item[i].price);
    }
    printf("Shipping Rs. %.2f\n",ShippingAmount);
    printf("Total Amount Rs. %f", totalAmount);
}
```